

RENJIE LIU

Shenzhen, China

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EDUCATION

Southern University of Science and Technology

Sep. 2023 – Jun. 2026

*M.phil. in Computer Science — Advised by **Prof. Bo Tang***

ShenZhen, China

- GPA: 3.62/4.00 (90/100)

Southern University of Science and Technology

Sep. 2019 – Jun. 2023

*B.Eng. in Computer Science — Advised by **Prof. Bo Tang** & **Prof. Xiao Yan***

ShenZhen, China

- GPA: 3.82/4.00 (91/100)
- Key Coursework: Linear Algebra (93), Discrete Math (96), Algorithm Design & Analysis (97), Database Principles (91), Computer Organization (94), Computer Network (96), Artificial Intelligence (92)
- Language Proficiency: TOEFL (104), CET-6 (590)

EXPERIENCE

NYU System Group, New York University

Dec. 2024 – Jun. 2025

*Student Intern — Advised by **Prof. Jinyang Li***

- General and effective **GraphRAG** solutions and robust **Agent** program development.

Shanghai AI Lab, Amazon Web Services

Jun. 2022 – Nov. 2024

*Applied Scientist Intern in **DGL Team** — Advised by **Dr. Minjie Wang***

- Efficient **GNN training** on large-scale datasets in single-machine and distributed environments.

PUBLICATIONS

* **PolyG: Effective and Efficient GraphRAG with Adaptive Graph Traversal**

Renjie Liu, Haitian Jiang, Xiao Yan, Bo Tang, and Jinyang Li

<https://arxiv.org/pdf/2504.02112>

* **DiskGNN: Bridging I/O Efficiency and Model Accuracy for Out-of-Core GNN Training**

Renjie Liu*, Yichuan Wang*, Xiao Yan, Zhenkun Cai, Minjie Wang, Haitian Jiang, Bo Tang, and Jinyang Li

ACM Conference on Management of Data (SIGMOD 2025)

*Equal Contribution

* **APT: Adaptive Parallel Training for Graph Neural Networks**

Kaihao Ma*, **Renjie Liu***, Zhenkun Cai, Xiao Yan, Xiang Song, Minjie Wang, and James Cheng

ACM Symposium on Principles and Practice of Parallel Programming (PPoPP 2025)

*Equal Contribution

* **Forming Scalable, Convergent GNN Layers that Minimize a Sampling-Based Energy**

Haitian Jiang, **Renjie Liu**, Xiao Yan, Zengfeng Huang, Zhenkun Cai, Minjie Wang, and David Wipf

International Conference on Learning Representations (ICLR 2025)

* **GSampler: General and Efficient GPU-based Graph Sampling for Graph Learning**

Ping Gong, **Renjie Liu**, Zunyao Mao, Zhenkun Cai, Xiao Yan, Cheng Li, Minjie Wang, and Zhuozhao Li

ACM Symposium on Operating Systems Principles (SOSP 2023)

PROJECTS

Towards Reliable and Transparent Web Agent

May 2025 – Present

Research Project Python

- Exploring a new execution paradigm for web agent by modularizing general REACT agents into co-operative components and regulating agent's action by explicit query planning with action templates.
 - ⇒ Observe that existing agents generally follow the REACT paradigm, which is prone to failure and difficult to debug.
 - ⇒ Exploring methods to modularize the agent's execution with improved reliability and transparency.

Effective and Efficient Graph RAG Solution

Aug. 2024 – May 2025

Research Project (PolyG) Python

- Propose a new benchmark tailored for Graph RAG and a new algorithmic solution that is both effective and efficient with adaptive graph traversal.
 - ⇒ Observe that existing GraphQA benchmarks are biased towards one type of graph queries.
 - ⇒ Propose a benchmark having a comprehensive set of query patterns of varying complexities.
 - ⇒ Propose a Graph RAG solution that adaptively selects the best traversal methods to retrieve the desired context.

Efficient out-of-core GNN Training on Large-scale Graphs 🌀

Sep. 2023 – July 2024

Research Project (DiskGNN) Python, C++, CUDA

- Build a system which achieves high I/O efficiency and thus fast training without hurting model accuracy for out-of-core GNN training where the whole graph can not fit in CPU memory.
 - ⇒ Observing that offline sampling scheme does not degrade model accuracy by sampling mini-batches beforehand.
 - ⇒ Propose a suite of designs based on offline sampling, including four-level feature store and batched feature packing.

Adaptive Parallel Training for Graph Neural Networks 🌀

April 2023 – Dec. 2023

Research Project (APT) Python, C++, CUDA

- Build a general framework capable of automatically determine the best parallelism for distributed GNN training based on hardware environments and models.
 - ⇒ Provide a comprehensive survey of existing parallel GNN training schemes and summarize their trade-offs.
 - ⇒ Drive an abstraction for all parallel schemes from the perspective of matrix segmentation & multiplication.
 - ⇒ Design a cost model to automatically determine the best parallelism given specific environments and configurations.

General and Efficient GPU-based Graph Sampling for Graph Learning 🌀 Aug. 2022 – April 2023

Research Project (gSampler) Python, C++, CUDA

- Build a high-performance GPU sampling framework for GNN training that can express diverse sampling algorithms by sketching sampling process using matrix operations.
 - ⇒ Survey 15 popular sampling algorithms, and propose a general 4-step ECSF model to sketch the sampling process.
 - ⇒ Propose comprehensive matrix-centric APIs for graph sampling based on the ECSF model.
 - ⇒ Incorporate a suite of sampling-oriented optimizations, e.g., kernel fusion, format selection and batching.

HONORS AND AWARDS

- China National Scholarship Nov. 2024
- University Merit Scholarship for Outstanding Student: The 1st Class Sep. 2023, 2022
- Outstanding Graduates of Department of Computer Science and Engineering Jun. 2023
- ACM SIGMOD 2022 Programming Contest: World's Finalist April 2022

EXPERIENCES

Teaching Assistant

- **Computer Organization and Architecture** Mar. 2022 – Jun. 2022
 - Designed and graded course assignments and the course project (a Minisys CPU written in Verilog on FPGA).
 - Answered questions about CPU architecture, MIPS and Verilog programming in laboratory courses.
- **Introduction to Computer Programming** Sep. 2020 – Jun. 2021
 - Designed and graded course assignments and the course project (a Minesweeper implemented by Java).
 - Answered questions about programming skills and algorithms in laboratory classes.